Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Bateman, Ted A. Ph.D.		
Project Title:	Examination of Anti-Resorptive and Anabolic Treatments/Stimuli on Unloading Induced Osteoporosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27599	**Congressional District:**	4
Comments:	Previous affiliation was Clemson University; PI moved to UNC in fall 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	10
No. of Master's Candidates:	7	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Simske, Steven (University of Colorado) Norrdin, Robert (Colorado State University)		
Grant/Contract No.:	NCC 9-58-BL00403		
Performance Goal No.:			
Performance Goal Text:			

Studies examining the calcium balance during extended spaceflight on the Russian Space Station Mir and the International Space Station suggest that 1) both bone resorption and formation are altered to adversely affect bone density, and 2) the time required to return bone mass to pre-flight levels may be considerably longer than the period of microgravity exposure. Countermeasures to prevent the loss of bone mass, and therapies to more quickly replace bone post-flight, are desirable to not only prevent fractures and kidney stones during and shortly after long-duration missions, but to also prevent microgravity-associated bone loss from exacerbating naturally occurring osteoporosis. Given the severity of bone loss, the long-term health concerns, and the numerous biomedical problems associated with spaceflight, every effort should be made to develop countermeasures that require minimal intervention.
In addition to micro- and reduced-gravity, astronauts on exploratory missions will be exposed to a composite of solar and cosmic radiation. To date, the doses and dose-rates of space radiation were not thought to be large enough to affect bone. In addition to the countermeasure regimen development component of our NASA applicable work, we have expanded the scope of our lab to include examining the effects of space relevant sources of radiation, and have identified profound bone loss at doses that approach what astronauts will experience on the moon and Mars. This work is being performed by combining resources from an NSBRI Synergy grant (one year), Procter and Gamble Pharmaceuticals, South Carolina Space Grant Consortium, and NASA/SC EPSCoR grants in addition to using resources from this primary NSBRI grant. This is a significant additional focus, but NASA's Exploration Roadmap has also changed with an increased emphasis on lunar exploration challenges. The addition of space radiation-induced bone loss to our research fits this change in the Roadmap.

In the first year of this grant we developed a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Both bone mass and turnover rates are important for evaluating successful recovery. The significance of these factors will be illustrated by first demonstrating that low doses of anti-resorptive agents can prevent in-flight loss of bone mass and allow bone turnover to recover more quickly. Fundamental to this objective was the determination of the minimum, efficacious therapeutic doses of anti-resorptives with studies employing mice administered either osteoprotegerin (OPG) or zoledronate. Following this, approximately 500 µg/kg for OPG and 45 µg/kg for zoledronate were determined to be the minimum, efficacious comparable doses. A paper reporting this "proof of concept" approach to developing a minimum, efficacious dose has been submitted to the Journal of Applied Physiology.

Task Description:

For the second phase of the project these doses of OPG and zoledronate were tested for their ability to prevent disuse osteoporosis with the hindlimb suspension model in mice. Both antiresorptive therapies, at these lower doses, maintained bone mass, strength and structural properties at the level of fully loaded, placebo treated control mice. Additionally, no significant inhibition of bone formation, a common consequence of antiresorptive therapies was not observed. This study essentially proves the efficacy of minimum dosing, and demonstrates its potential for allowing bone formation and turnover to return to pre-flight levels as quickly as possible after flight. A paper is in preparation for the Journal of Applied Physiology.

For the space radiation-induced bone loss project, we have reported a profound loss of bone mass at a dose, 2 Gray (Gy), lower than previously reported, approaching levels of concern for exploratory missions. Significant losses in trabecular volume fraction were observed for all radiation species four months after exposure: gamma, (-29%), proton, (-35%), carbon (-39%), and iron (-34%). Trabecular connectivity density, thickness, spacing, and number were also affected. These data have clear implications for clinical radiotherapy, in that bone loss in an animal model has been demonstrated at low doses. Additionally, these data suggest that space radiation has the potential to exacerbate the bone loss caused by microgravity, though lower doses and dose rates need to be studied.

The Synergy grant from NSBRI (small, one year grant) funded our examination of the dose threshold for bone loss caused by proton radiation (modeling a solar flair), examining doses of 2, 1 and 0.5 Gy. This project concluded that the threshold for significant bone loss is between 0.5 and 1 Gy, a dose astronauts may possibly receive during a lunar mission and are likely to receive on a trip to Mars. This primary NSBRI grant (in addition to the other small, short-term grants received from P&G, SC Space Grant and EPSCoR) have been used to analyze bones sent to us from NASA radiation biology investigators from around the country. These bones, collected as secondary tissue from other NASA projects, are allowing us to efficiently further define the potential problem of bone loss caused by space radiation.

From this work we have been able to make several conclusions regarding the functional loss of bone caused by space-relevant sources of radiation. 1) Long-term bone loss from iron radiation is a result of local, regional, and systemic factors. 2) Analysis of serum from these iron irradiated rats suggest a long-term increase in bone turnover. 3)Finally, dose rate is an important contributing variable in the degree of bone loss for low-LET radiation. Mice exposed to 2 Gy of protons over 36 hours lost approximately half the bone of mice exposed to the same acute dose.

Rationale for HRP Directed Research:

A recent retrospective analysis of more than 6,000 postmenopausal women receiving radiotherapy for cervical, rectal, and anal cancers established a significant correlation between radiation therapy and increased fracture risk (Baxter et al., JAMA, 2005). Risk of hip fracture among these patients was significantly elevated five years post-therapy (relative risks of 1.66, 1.65 and 3.16, respectively).
Skeletal complications from radiation therapy have been described for breast, brain, and pelvic cancers and leukemia (bone-marrow transplants). These problems include reduced bone density, fractures, osteoradionecrosis and back pain affecting both children and adults. Improved survivorship rates of cancer patients receiving radiotherapy increase the importance of understanding the mechanisms and long-term effects of radiation-induced bone loss.

Radiation therapy is administered as whole-body (preceding bone marrow transplant) or local, targeting the tumor. Protocols vary considerably, but generally whole-body irradiation totals 10-15 Gy and is administered in daily fractions of 1-2 Gy, whereas local therapy totals 40-70 Gy delivered in approximately 2 Gy fractions.

Research Impact/Earth Benefits:

Understanding the role of radiation therapy in fracture risk is complicated by the fact that this bone loss is often accompanied by several contributors to low bone density, including premature menopause, androgen deficiency, and/or chemotherapy and the direct effect of cancer on bone loss. Identifying the independent effects of radiation on the skeletal system requires an animal model. To date, one does not exist. Further development of this model is necessary, including the characterization of the time course of the loss, the potential sparing effect of fractionation and low-dose rate, and identification of dose thresholds, so that molecular mechanisms and therapies can be examined.

	The target audience for the NSBRI funded research is interested in applied bone biomechanics and physiology. The research journals we target are the Journal of Orthopaedic Research, Journal of Biomechanics, Journal of Bone and Mineral Research, Bone and the Journal of Applied Physiology.
Task Progress:	The determination of the minimum, efficacious doses of OPG and zoledronate was obtained through a series of iterative studies, with the dose yielding maximal increases in trabecular density and/or connectivity in one study serving as the medium dose for the next study. With each iteration the bracketing dose range on the medium dose was narrowed. Minimum efficacious doses of 500 ug/kg for OPG and 45 ug/kg for zoledronate were determined to be comparable. These doses were tested with the hindlimb suspension model in mice and maintained bone mass and strength in skeletally unloaded mice to those of placebo treated loaded mice. For the radiation project, we have been able to make several conclusions regarding the functional loss of bone caused by space-relevant sources of radiation, and have developed mechanistic hypotheses. 1) Long-term (nine months after exposure) bone loss from iron radiation is a result of local, regional, and systemic factors. We have identified bone loss in hind limbs from rats where only the torso was irradiated. Additionally, there is greater bone loss in bone closer to regions that received radiation. 2) Analysis of serum from these iron irradiated rats suggest a long-term increase in bone turnover, with TRAP, osteocalcin, and alkaline phosphotase all tending to be higher in irradiated rats than non-irradiated controls nine months after exposure. 3) We have identified a very rapid and large bone loss in mice receiving high doses (7 Gy) of gamma radiation: 55% of trabecular bone lost in two weeks. The implications of this conclusion is more relevant to clinical applications of radiation, because of the high dose, but it allows us to clearly identify an increase in bone resorption, rather than solely an inhibition of bone formation, as a primary cause for the bone loss. 4) Dose rate is an important contributing variable in the degree of bone loss for low-LET radiation. Mice exposed to 2 Gy of protons over 36 hours lost approximately half the bone of mice exposed to the same acute dose (administered over minutes). 5) Analysis of the dose-rate study is ongoing. We can tentatively conclude that the dose threshold for bone loss from proton radiation is between 0.5 and 1 Gy. Mechanistic Hypothesis: We hypothesize that radiation-induced bone loss is caused by an increase in osteoclastic bone resorption mediated by reactive oxygen species and proinflammatory-cytokine activation of RANKL expression, and that trabecular bone is particularly sensitive due to its large surface area interface with hematopoietic tissue.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Articles in Peer-reviewed Journals	Hamilton SA, Pecaut MJ, Gridley DS, Travis ND, Bandstra ER, Willey JS, Nelson GA, Bateman TA. "A murine model for bone loss from therapeutic and space-relevant sources of radiation." J Appl Physiol. Epub 2006 Jun 1. PMID: 16741258 , Jun-2006
Articles in Peer-reviewed Journals	Lloyd SA, Bateman TA. "Minimum, comparable efficacious doses of zoledronic acid and osteoprotegerin: maximizing bone quality." J Appl Physiol. Submitted for Publication, 2006 August. , Aug-2006
Articles in Peer-reviewed Journals	Willey JS, Grilly LG, Howars SH, Hamilton SA, Pecaut MJ, Obenaus A, Gridley DA, Nelson G. "Effects of 56Fe Radiation on the Architecture and Structural Properties." Submitted for Publication, Radiation Research, June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Yuan YY, Lau AG, Kostenuik PJ, Morony S, Adamu S, Asuncion F, Bateman TA. "Soluble RANKL has detrimental effects on cortical and trabecular bone volume, mineralization and bone strength in mice." Bone. Submitted for Publication, 2006 August. , Aug-2006
Awards	Willey J. "Jeff Willey: Seminar Speaker First Place, Clemson University Department of Bioengineering Fall Seminar Series. Radiation and bone loss: establishing a hypothesis for the molecular and cellular mechanisms. December 2005." Dec-2005
Awards	Hamilton S. "Sarah Hamilton: Travel Award for the 52nd Annual Meeting of the Radiation Research Society, Denver, CO, October 16-19, 2005." Oct-2005
Awards	Lloyd S. "Shane Lloyd: First Place Undergraduate Poster Competition and Travel Grant, 2005 Annual Meeting of the American Society for Gravitational and Space Biology, Reno, NV, November 1-4, 2005." Nov-2005
Awards	Lu T, Anderson E, Tirado L, Riffle S. "Teng Lu, Erica Anderson, Leidamarie Tirado and Stephanie Riffle: Undergraduate Research Program Awards, Howard Hughes Medical Institute/SC LIFE, Clemson University 2006-2007 academic year." Sep-2006
Awards	Yuan Y. "Yuyu Yuan: Webster Jee Young Investigator Award at the 2nd International Conference on Osteoporosis and Bone Research, Chengdu, China, October 19-23, 2005." Oct-2005